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Differential Shoreline Dynamics of Inhabited and Uninhabited Estuarine Islands in the Indian Sundarb - 6/29/2026

Differential Shoreline Dynamics of Inhabited and Uninhabited Estuarine Islands in the Indian Sundarbans (1990–2024)

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Abstract

Rapid relative sea-level rise (RSLR), strong tidal forcing, sediment-supply disturbance and frequent cyclones have severely affected Indian Sundarbans, which is the world’s largest continuous mangrove delta (Hazra et al., 2002; Pethick & Orford, 2013; IPCC, 2021). To understand how human settlement and embankment affect the shoreline behaviour we have chosen a paired-island design (inhabited vs uninhabited). Kumirmari (19.09 km², inhabited–interior), Hamilton (84.94 km², uninhabited–interior), Mousuni (26.19 km², inhabited–coastal) and Lothian (36.30 km², uninhabited–coastal) were chosen for analysis.

To extract shoreline, Multi-temporal Landsat archive (1990–2024) has been processed in Google Earth Engine (GEE) along with thresholding using MNDWI + Otsu (McFeeters, 1996; Xu, 2006; Otsu, 1979; Gorelick et al., 2017). Using FES2022 global tide model each shoreline is tidally corrected (Lyard et al., 2021; CNES, 2024). Shoreline change was quantified using EPR and LRR along DSAS-equivalent transects (Thieler et al., 2009; Himmelstoss et al., 2018), Cyclone-induced inundation was assessed using Sentinel-1 SAR backscatter (Torres et al., 2012; Lee, 1980). Future shoreline positions (2034, 2044) were projected using a 2-D Kalman filter (Kalman, 1960; Long & Plant, 2012; Vitousek et al., 2017). Results show that the inhabited-coastal islands are the most vulnerable: Mousuni $\sim 72\%$ transect net erosional and the most eroded transect (-24.99 m yr^{-1}) within $\sim 6\%$ of the highest reported Sundarbans–Hooghly erosion rate ($\sim -23.5 \text{ m yr}^{-1}$; Bandyopadhyay, 1997; Gopinath & Seralathan, 2005). Mousuni shows a $\sim 93 \text{ m}$ retreat ($\sim 84\%$ transect eroding) according to Kalman forecast of 2044. In contrast, sheltered interior islands exhibit near-stable and statistically indistinguishable shoreline behaviour: Kumirmari (-0.62 m yr^{-1}) and Hamilton (-0.89 m yr^{-1}). Lothian remains stable to slightly accretional ($+0.54 \text{ m yr}^{-1}$, LRR $+1.44$). Habitation and erosion have strong impact in the open coast but in the sheltered interior it is negligible: Island setting $\times$ habitation interaction ($+2.10 \text{ m yr}^{-1}$, 95% CI [$+0.25$, $+4.08$]). These findings have direct implications for coastal management, embankment policy and climate adaptation planning.

Keywords: Indian Sundarbans, shoreline change, DSAS, EPR, LRR, MNDWI, FES2022 tidal correction, Sentinel-1 SAR, Kalman filter.

1. Introduction

Low-lying deltaic coasts are among the most vulnerable environments worldwide (Syvitski et al., 2009; Ericson et al., 2006; Tessler et al., 2015). Indian and Bangladeshi Sundarbans which are present in Ganga-Brahmaputra-Meghna (GBM) delta is a classic example of such environment. It is affected by multiple factors such as: mangrove ecology, intense human pressure, macro-tidal regime and sediment dynamics (Giri et al., 2011; Ghosh et al., 2015; Woodroffe et al., 2016). The low-lying tropical deltas will be dominantly driven by relative sea-level rise in the 21st Century according to the IPCC AR6 and due to subsidence and sediment-starvation the Sundarbans region has experienced rates of relative sea-level rise substantially higher than the global mean (Pethick & Orford, 2013; Hazra et al., 2002). The net result of all these processes is captured by the Shoreline, which acts as a geo-indicator (Boak & Turner, 2005; Crowell et al., 1991). In regards to coastal-hazard assessment, climate adaptation and erosion management, the historical movement of the shoreline needs to be quantified. Sagar, Ghoramara, Namkhana and other Hooghly-estuary islands follow a long legacy of shoreline change study in the Indian Sundarbans (Bandyopadhyay, 1997; Gopinath & Seralathan, 2005; Ghosh et al., 2001; Bandyopadhyay et al., 2014; Rahman et al., 2011). However, two critical research gaps remain. First, most studies focus on either a single island or the entire biosphere, failing to systematically isolate how human settlement and engineered embankments alter shoreline behavior.

Second, very few studies focus on the correction of tidal stage, though apparent change can be there in uncorrected instantaneous waterline in macro-tidal Sundarbans just because of tide (Boak & Turner, 2005; Lyard et al., 2021).

A 2×2 factorial design was adopted, defined by two factors: (i) settlement status (inhabited vs uninhabited) and (ii) geomorphic position (interior vs coastal/seaward), with each combination represented by one island. This design allows an interaction effect to be assessed — that is, whether the influence of settlement increases with coastal exposure.

To obtain robust conclusions, we employed and integrated four remote-sensing approaches: (i) GEE-based MNDWI + Otsu shoreline extraction (McFeeters, 1996; Xu, 2006; Otsu, 1979; Gorelick et al., 2017); (ii) per-epoch FES2022 tidal correction (Lyard et al., 2021; CNES, 2024); (iii) DSAS-equivalent transect statistics (EPR, LRR) for change rates (Thieler et al., 2009; Himmelstoss et al., 2018; Dolan et al., 1991); and (iv) a 2-D Kalman filter for shoreline forecasting (Long & Plant, 2012; Vitousek et al., 2017).

In addition, Sentinel-1 SAR was used to observe cyclone-period (Amphan 2020, Yaas 2021, Remal 2024) inundation and backscatter change (Torres et al., 2012; Lee, 1980).

This study has three specific objectives: (1) to produce and quantify tidally corrected shoreline-evolution maps for the period 1990–2024; (2) to statistically compare change rates between inhabited and uninhabited islands and between interior and coastal settings; and (3) to forecast mid-century (2034, 2044) shoreline positions and identify the most vulnerable locations using a Kalman filter.

We hypothesise that the inhabited-coastal setting (Mousuni) will show the highest erosion rate, owing to the combined effects of embankment-induced sediment starvation, tidal prism disturbance and direct marine exposure (Bandyopadhyay, 1997; Pethick & Orford, 2013).

2. Study Area

The location and physiographic setting of the four study islands are shown in Figure 1.

Figure 1

Study Area and Location of the Four Study Islands in the Indian Sundarbans (South 24 Parganas, West Bengal)

Note.
Inset: position within the Ganga–Brahmaputra–Meghna delta.
Islands: Kumirmari and Hamilton (interior), Mousuni and Lothian (coastal).

Table 1.
Study islands and the 2×2 paired design (areas from the GEE MNDWI 2024 reference).

Inhabited (settled, embanked)

Uninhabited (protected)

Interior (estuarine)

Kumirmari — 19.09 km² (Gosaba Block)

Hamilton — 84.94 km²

Coastal (seaward)

Mousuni — 26.19 km²

Lothian — 36.30 km² (Wildlife Sanctuary)

3. Data and Methods

The overall methodology adopted in this study is summarised in Figure 2.

Figure 2

Methodological Workflow of the Study

Note.

Shoreline has been extracted using MNDWI from Landsat 5,7,8,9 imagery and Otsu thresholding (McFeeters, 1996; Xu, 2006; Otsu, 1979; Gorelick et al., 2017) and it has been tidally corrected using FES2022 ocean-tide model (Lyard et al., 2021; CNES, 2024), in a parallel Sentinel-1 SAR branch the cyclone-period inundation is captured (Torres et al., 2012; Lee, 1980), Using DSAS-equivalent transect the change rate has been computed (Thieler et al., 2009; Himmelstoss et al., 2018; Dolan et al., 1991) and furthermore the future shoreline has been projected using 2-D Kalman filter (Kalman, 1960; Long & Plant, 2012; Vitousek et al., 2017).

3.1 Satellite data and cloud processing

For optical analysis Landsat Collection-2 Surface Reflectance archive (Landsat 5 TM, 7 ETM+, 8-9 OLI) has been used which has been accessed and processed in Google Earth Engine (GEE) (Gorelick et al., 2017; Roy et al., 2014; Wulder et al., 2019).

Twelve epochs (1990, 1995, 2000, 2005, 2008, 2010, 2013, 2016, 2019, 2020, 2022, 2024) has been chosen so that it covers both the long term and recent cyclone-year.

For each epoch, dry-season (pre-monsoon) image cloud-mask was used to create a median composite in order to minimise residual cloud and sensor-noise.

For SAR component, Sentinel-1 GRD (C-band, VV/VH) imagery around the cyclone window was used (Torres et al., 2012).

Coordinate system for the whole was WGS 84/UTM Zone 45N (EPSG:32645).

3.2 Shoreline extraction: MNDWI + Otsu

To extract the land–water boundary, we used a single water index.

Water indices combine spectral bands to enhance the contrast between land and water surfaces.

The original index, NDWI, uses the green and near-infrared (NIR) bands (McFeeters, 1996):

$$\text{NDWI} = (\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR}) \quad (1)$$

The waters of the Sundarbans are turbid and sediment-laden, and NDWI may therefore misclassify turbid water and bright sediment surfaces.

To address this, we used the modified NDWI (MNDWI), which replaces the NIR band with the shortwave-infrared (SWIR) band and separates land from water more reliably (Xu, 2006; cf.

Gao, 1996):

$$\text{MNDWI} = (\text{Green} - \text{SWIR}) / (\text{Green} + \text{SWIR}) \quad (2)$$

MNDWI is high over open water and low over land.

Rather than selecting the land–water threshold by visual inspection, we applied the Otsu (1979) method, which determines it automatically.

The Otsu method evaluates every possible threshold and divides the pixels into two classes (water and land), with class probabilities ω_0 , ω_1 and mean index values μ_0 , μ_1 :

$$\omega_0(t) = \sum p(i) \text{ for } i \leq t \quad ; \quad \omega_1(t) = \sum p(i) \text{ for } i > t \quad (3)$$

$$\mu_0(t) = [\sum i \cdot p(i), i \leq t] / \omega_0 \quad ; \quad \mu_1(t) = [\sum i \cdot p(i), i > t] / \omega_1 \quad (4)$$

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FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.